Computer Applications in Business

Lecture 3 & 4
Interacting with Personal Computer & Ergonomics

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Devices used to interact with a Computer:

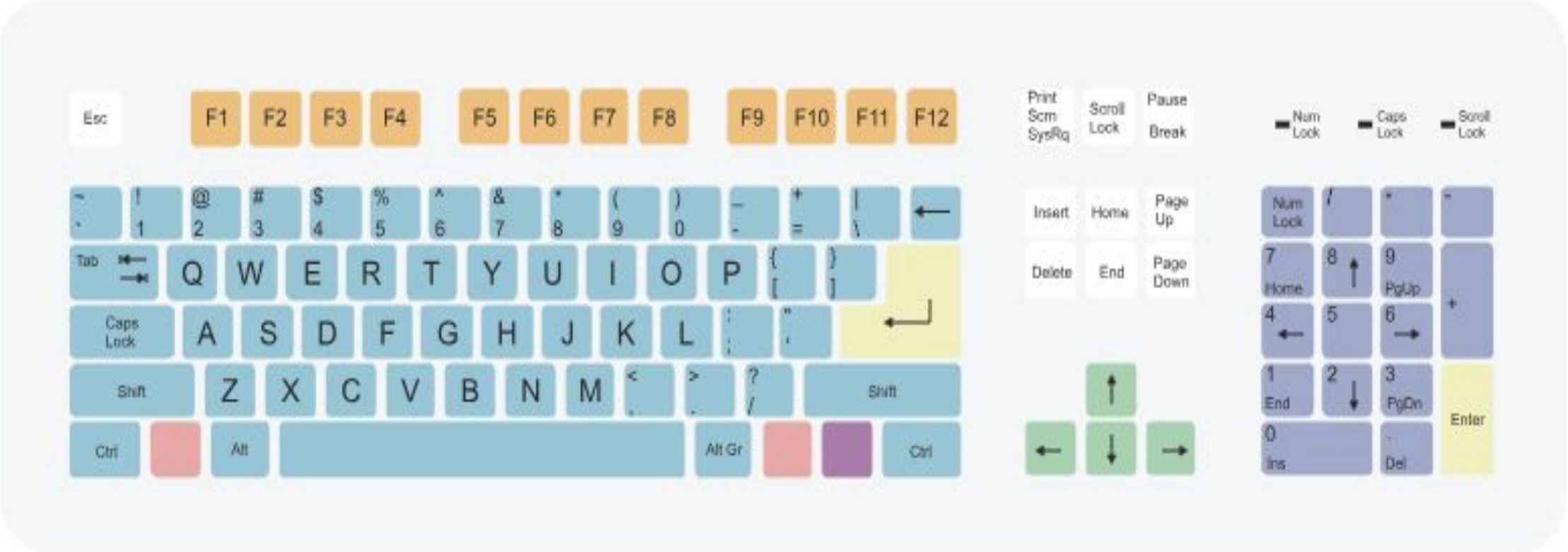
- Keyboard
- Mouse
- Touchpad

If you think of the CPU as a computer's brain, then you might think of the input devices as its sensory organs—the eyes, ears, and fingers. From the user's point of view, input devices are just as important as the CPU, perhaps even more important.

The Standard Keyboard Layout

A standard keyboard includes about 100 keys; each key sends a different signal to the CPU.
The keyboard was one of the first peripherals to be used with computers, and it is still the primary input device for entering text and numbers.

The standard keyboard has from 101 to 103 keys. The letters are in the Querty layout. Numbers in a row above, spacebar front and center.



Typewriter keys

Function keys

Enter keys

System keys

Numeric keypad

Other

Application key

Cursor control keys

The Alphanumeric Keys:

- The alphanumeric keys the area of the keyboard that looks like a typewriter's keys are arranged the same way on almost every keyboard. Sometimes this common arrangement is called the QWERTY (pronounced KWER-tee) layout because the first six keys on the top row of letters are Q, W, E, R, T, and Y.

The Modifier Keys:

- The SHIFT, ALT (Alternate), and CTRL (Control) keys are called modifier keys because they modify the input of other keys.

The Numeric Keypad :

The numeric keypad looks like a calculator's keypad, with its 10 digits and mathematical operators (+, -, *, and /). The numeric keypad also features a NUM LOCK key.

The Function Keys:

The function keys, which are labeled F I, fl, and so on. menus or dialog boxes. Each function key's purpose depends on the program you are using.

The Cursor-Movement Keys :

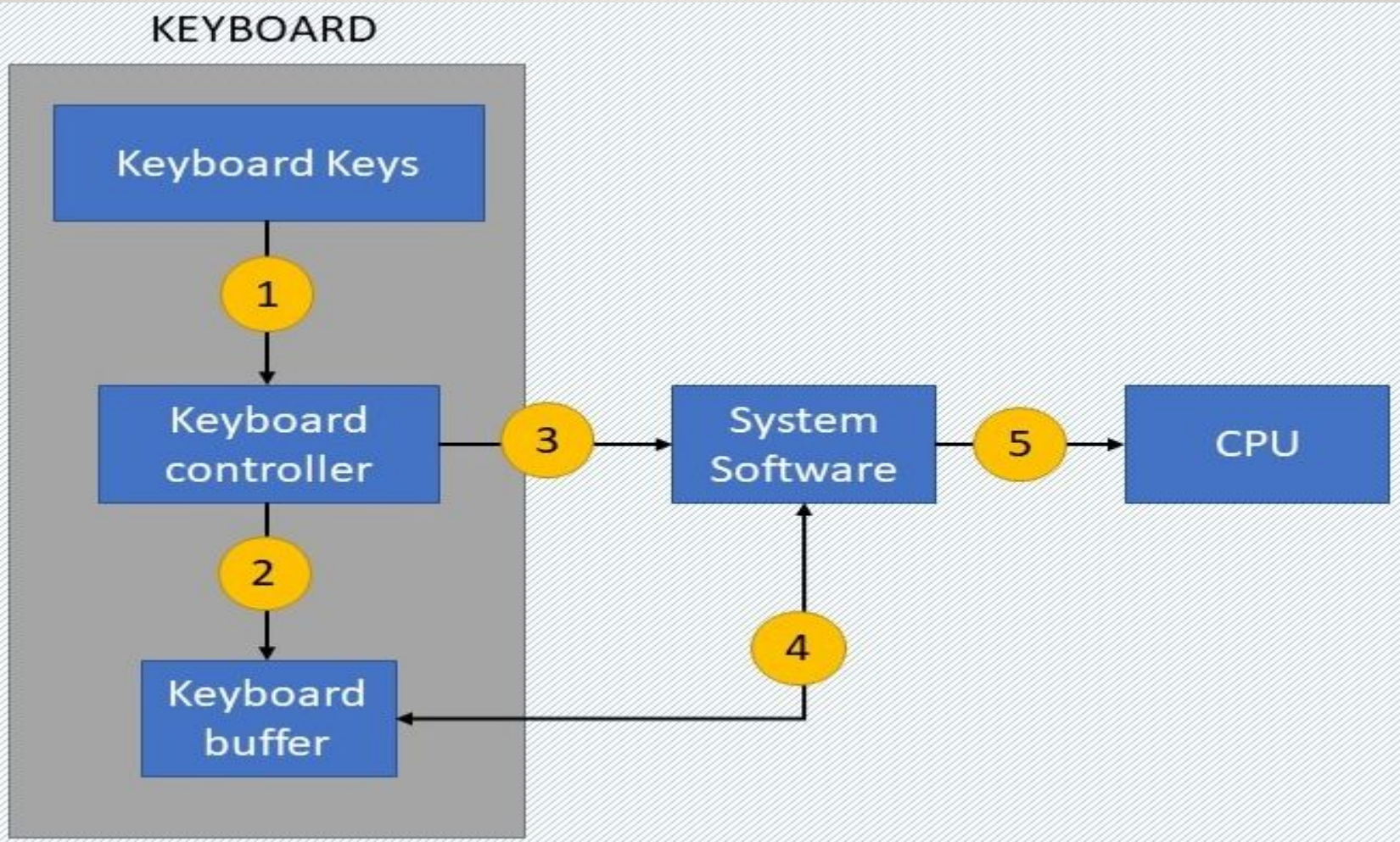
Most standard keyboards also include a set of cursor movement keys, which let you move around the screen without using a mouse. In many programs and operating systems, a mark on the screen indicates where the characters you type will be entered.

Special-Purpose Keys :

In addition to the five groups of keys described earlier, all IBM compatible keyboards feature six special-purpose keys, each of which performs a unique function.

- **Start**
- **shortcut**

How the Computer Accepts Input from the Keyboard



Using the Mouse

You use a mouse to move the pointer to a location on the screen, a process called pointing. Everything you do with a mouse is accomplished by combining pointing with these techniques:

» Clicking
» Double-clicking
» Dragging
» Right-clicking

Ergonomics and Input Devices

- **Ergonomics,** which is the study of the physical relationship between people and their tools—such as computers—addresses these issues.
- The term ergonomically correct means that a tool or a workplace is designed to work properly with the human body, and thus reduces the risk of strain and injuries.

Repetitive Stress Injuries: These ailments are called repetitive stress injuries (RSIs) or repetitive strain injuries and result from continuously using the body in ways it was not designed to work.

Here are some other tips that can help you avoid RSIs while working with your keyboard and mouse:

1. **Use an Ergonomic Keyboard :** An ergonomic keyboard allows you to hold your hands in a more natural position (with wrists straight, rather than angled outward) while typing.

2. Use a Padded Wrist Support :

If you type a lot, a wrist support can be helpful by allowing you to rest your hands comfortably when you are not actually typing.

3. Keep Your Wrists Straight :

When typing, your hands should be in a straight line with your forearms, when viewed either from above or from the side.

4. Sit Up Straight : Avoid crossing your legs in front of you or under your chair for long periods.

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5.Learn to type : You will use the keyboard more efficiently and naturally if you know how to type.

6. Take Frequent Breaks : Get up and move around for a few minutes each hour, and stretch occasionally throughout the day.

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Ergonomics and Monitor

Just as too much keyboarding or improper typing technique can lead to hand or wrist injuries, too much time at a monitor can endanger your eyesight. Protecting your eyesight means choosing the right kind of monitor and using it correctly.

Monitor tips:

- Adjust the monitor height so that the top of the screen is at—or slightly below—eye level.
- Your eyes should look slightly downward when viewing the middle of the screen.
- Position the monitor at least 20 inches (51 cm) from your eyes—about an arm's length distance. If your screen is larger, add more viewing distance.

- Keep glare down by adjusting the screen position.
- Maintain the distance between your screen-scanning eyes and your screen by tilting the monitor back 10° to 20°.
- If you wear bifocals (we'll never tell!), lower the monitor below eye level and turn your screen upward, tilting it back 30° to 45°.

Most input devices arc designed to be used by hand. Even specialized devices like touch screens enable the user to interact with the system by using his or her fingertips. Unlike keyboards and mice, many of these input devices are highly intuitive and easy to use without special skills or training.

- **Pen-based systems:** You hold the pen in your hand and write on a special pad or directly on the screen. You also can use the pen as a pointing device, like a mouse, to select commands by tapping the screen.

- Pen-based computers are commonly used for data collection, where the touch of a pen might place a check in a box to indicate a par that must be ordered or a service that has been requested

Touch Screens: Touch screens accept input by allowing the user to place a fingertip directly on the computer screen, usually to make a selection from a menu of choices. Most touchscreen computers use sensors on the screen's surface to detect the touch of a finger, bur other touch screen technologies are in use, as well.

- Touch screens have become common in fast-food restaurants, department stores, drugstores, and supermarkets, where they are used for all kinds of purposes, from creating personalized greeting cards to selling lottery tickets.

Game Controllers: You may not think of a game controller as an input device, but it is. Personal computers are widely used as gaming platforms, challenging dedicated video game units like the Sony PlayStation

Optical Input Devices:

- Bar Code Reader: Bar code readers are one of the most widely used input devices. The most common type of bar code reader is the flatbed model, which is commonly found in supermarkets and department stores.

Workers for delivery services, such as FedEx, also use handheld bar code readers in the Held to identify packages

Video

- The quality of the images that a monitor can display is defined as much by the video card (also called the video controller or the video adapter) as by the Monitor itself.
- the video controller is an intermediary device between the CPU and the monitor. It contains the video-dedicated memory and other circuitry necessary to send information to the monitor for display on the screen.
- In most computers, the video card is a separate device that is plugged into the motherboard.
- The video controller contains its own on-board processor and memory, called video RAM

Sound

- The most complicated part of a computer’s sound system is the sound card.
- A computer's sound card is a circuit board that converts sound from analog to digital form, and vice versa, for recording or playback. A sound card actually has both input and output functions.
- your voice, for instance, you connect the microphone to the sound card’s input jack. Other audio input devices connect to the sound card as well, such as the computer's CD-ROM or DVD drive.
- The sound card translates digital signals into analog signals that drive the speakers.
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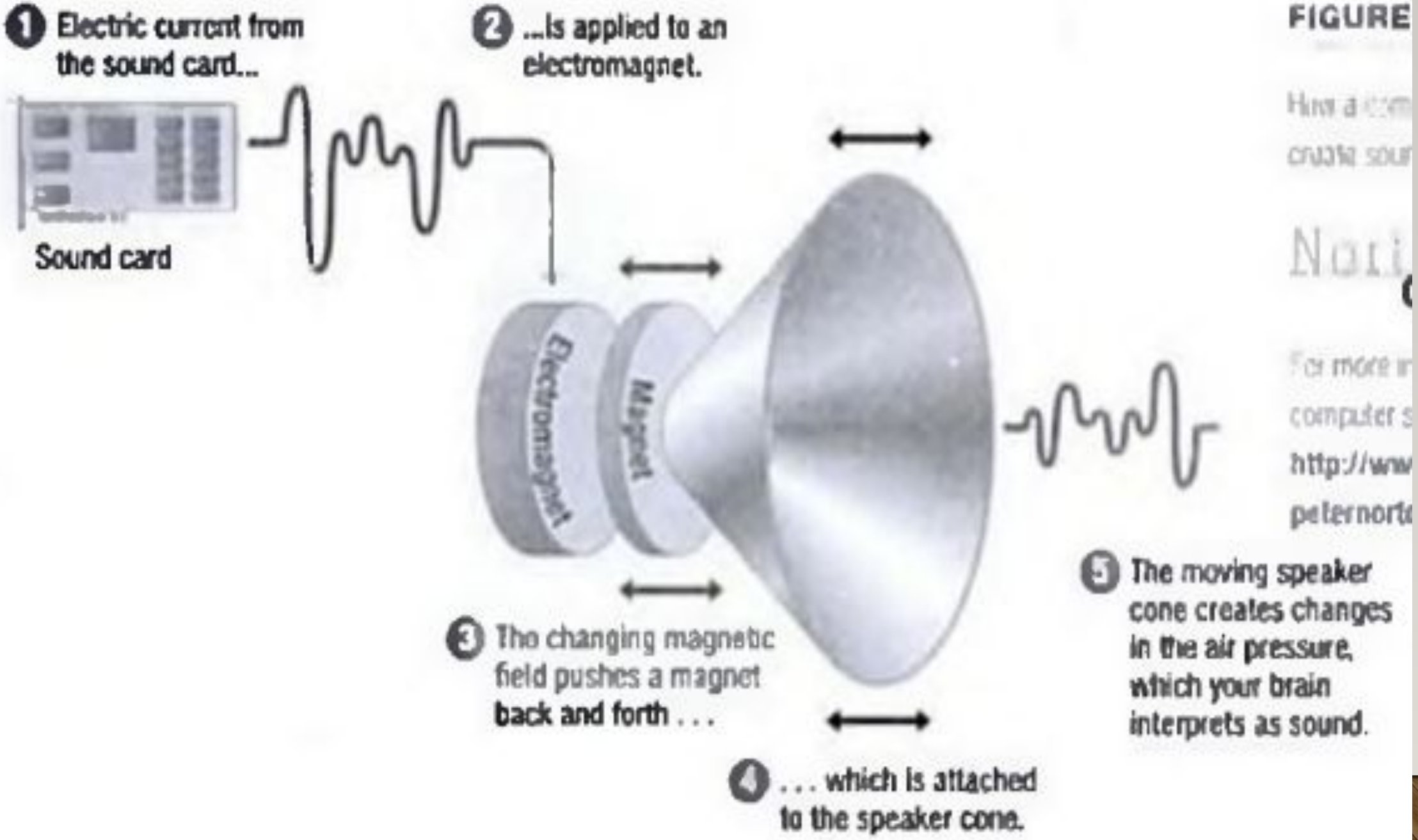

1 Electric current from the sound card...
2 ...is applied to an electromagnet.
Sound card
Electromagnet
Magnet
3 The changing magnetic field pushes a magnet back and forth . . .
4 . . . which is attached to the speaker cone.
5 The moving speaker cone creates changes in the air pressure, which your brain interprets as sound.

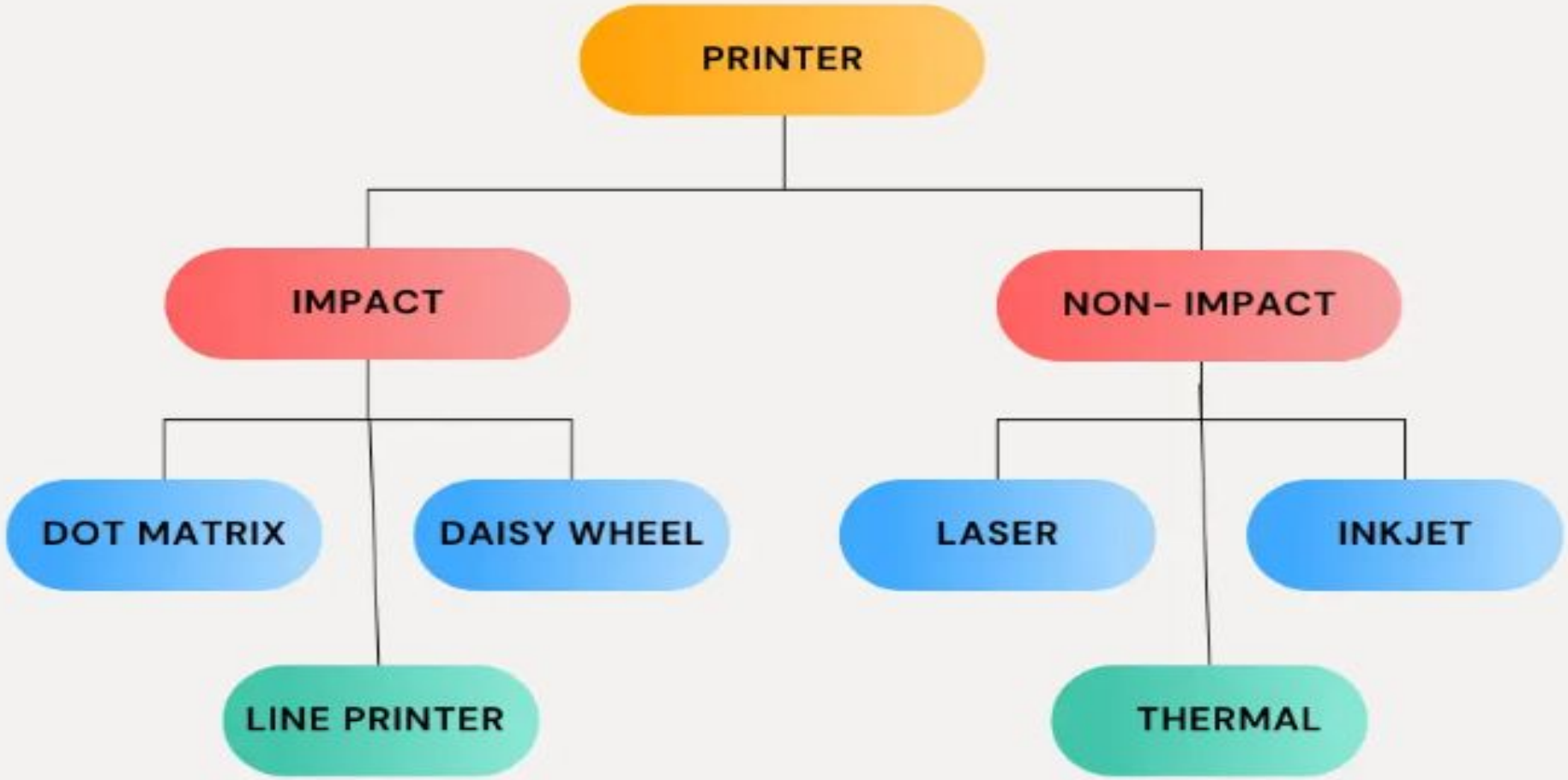
Printers

In the field of computing, a printer is considered a peripheral device that serves the purpose of creating a permanent representation of text or graphics, usually on paper.

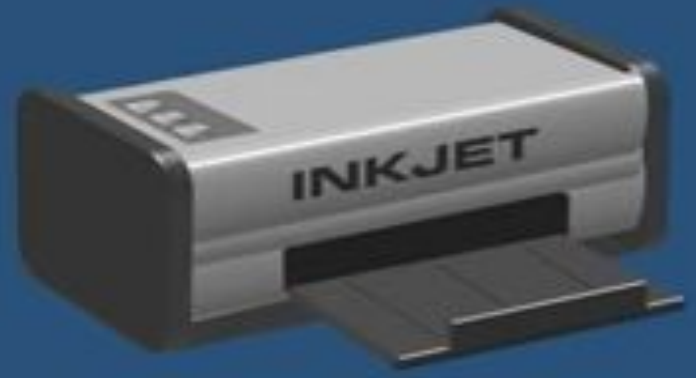
Printers allow the ability to print copies of documents or photos onto paper from a computer

- Printers fall into two general categories: impact and nonimpact.
- Impact printers create an image on paper by using a device to strike an inked ribbon, pressing ink from the ribbon onto the paper. Nonimpact printers use various methods to place ink on the page.

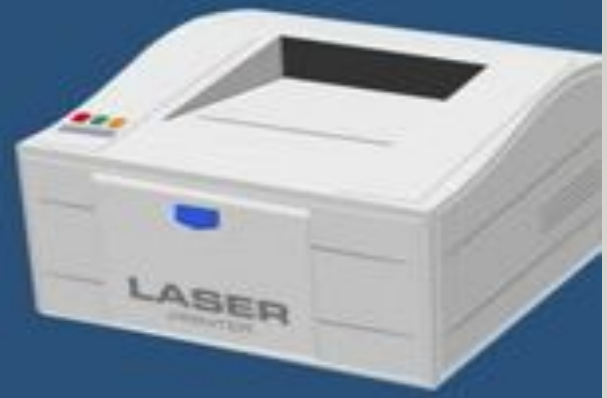
TYPES OF PRINTERS



Printers



MOST COMMON

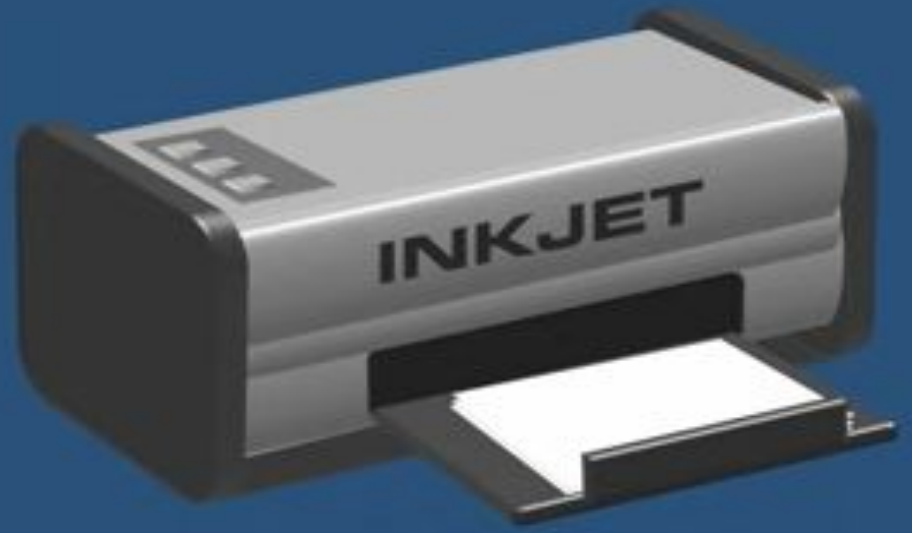


INKJET

LASER

LESS COMMON

Inkjet *PRINTER*

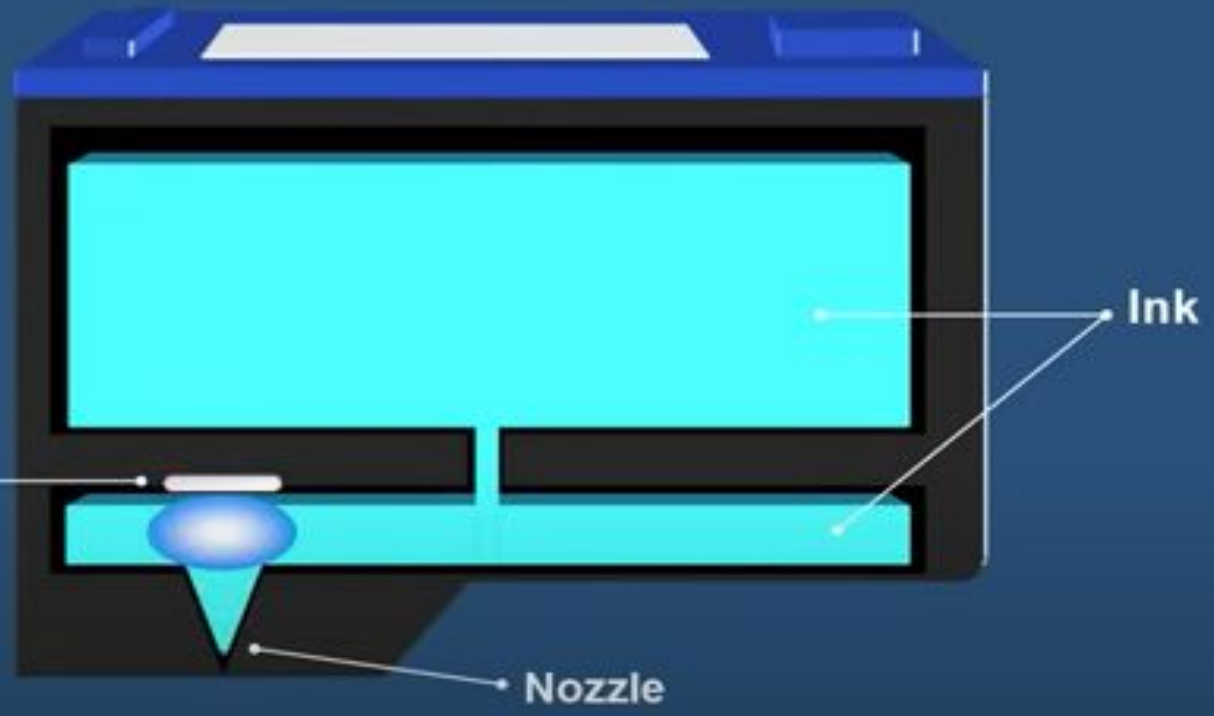


- The most common printers that are used in homes.
- Print head moves back and forth across the paper.
- Print head places ink on paper in very tiny dots.

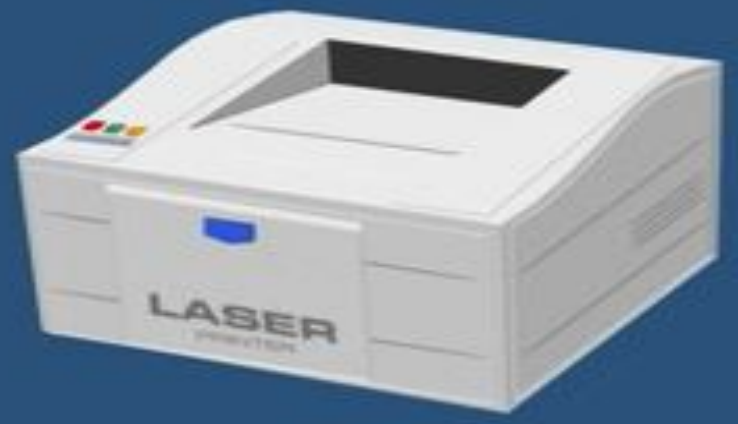
Inkjet *PRINTER*

Thermal Bubble Technology

Used in HP and Canon printers.



Laser *PRINTER*

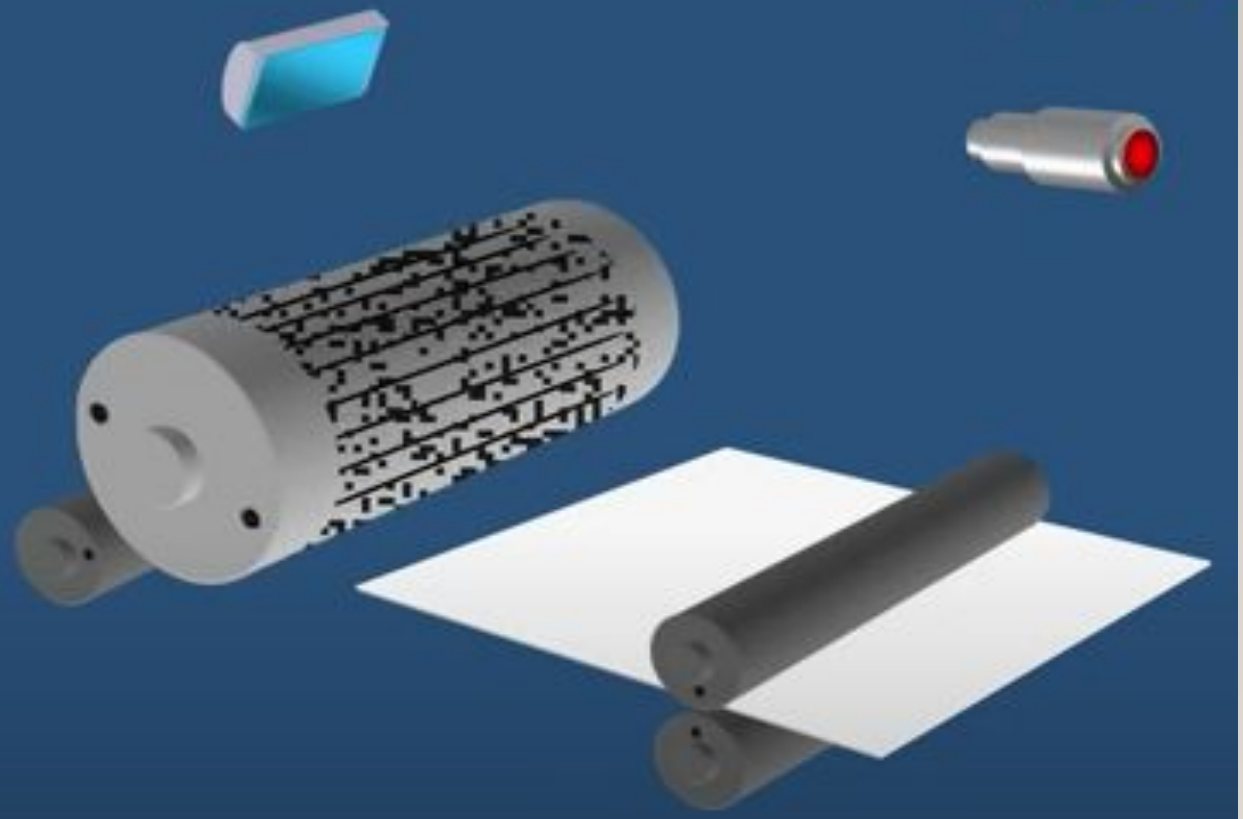


- Laser printers come in different sizes.
- From small personal ones, to larger ones for businesses.
- Provides the highest quality print.
- More expensive than inkjet printers.

Laser *PRINTER*

BASIC OPERATION

- Electric charge is placed on a rotating drum.
- Laser discharges a lower electrical charge on the drum.
- Laser draws the image that is going to be printed on the drum itself.
- Drum is coated with toner.
- Toner only clings to the areas where the laser has drawn.



Thermal *PRINTER*

Finished

Thermal
Paper

Prints by using heat.

Commonly used for printing labels and barcodes.

- A dot matrix printer is an impact printer. It uses a print head that contains a duster of pins. The printer can push the pins out in rapid sequence to form patterns. The pins are used to press an inked ribbon against paper, creating an image.
- An inkjet printer is an example of a nonimpact printer. It creates an image by spraying tiny droplets of ink
- Laser printers produce higher-quality print and are fast and convenient to use, but they are also more expensive than ink jet printers. Laser printers are available in both color and black and white
- Thermal-wax and dye-sublimation printers are used primarily by print shops and publishers to create high-quality color images.
- Photo printers are specialized printers used to print color photographs taken with digital cameras.